



Harvard John A. Paulson
School of Engineering
and Applied Sciences

Explain Yourself!

Leveraging Language Models

for Commonsense Reasoning

Authors: Nazneen Fatema Rajani, Bryan McCann, Caiming Xiong,
Richard Socher

Presenters: Karly Hou, Eshika Saxena, Leonard Tang, Kat Zhang

Introduction

- **Commonsense reasoning**: making human-like presumptions and judgements about ordinary situations
- Modern ML methods struggle with commonsense reasoning
- Explanations help verbalize reasoning that models learn while training
- Common sense Question Answering (CQA) dataset

Question: While eating a **hamburger with friends**,
what are people trying to do?
Choices: **have fun**, tasty, or indigestion

How do these models perform reasoning and to what extent is that reasoning based on world knowledge?

Key Contributions

Common Sense Explanations (CoS-E): Collected human explanations (annotations and natural language explanations) to build on top of CQA

Question:	While eating a hamburger with friends , what are people trying to do?
Choices:	have fun , tasty, or indigestion
CoS-E:	Usually a hamburger with friends indicates a good time.
Question:	After getting drunk people couldn't understand him, it was because of his what?
Choices:	lower standards, slurred speech , or falling down
CoS-E:	People who are drunk have difficulty speaking.
Question:	People do what during their time off from work ?
Choices:	take trips , brow shorter, or become hysterical
CoS-E:	People usually do something relaxing, such as taking trips, when they don't need to work.

Table 1: Examples from our CoS-E dataset.

Key Contributions

1. Common Sense Explanations (CoS-E)
2. Commonsense Auto-Generated Explanations (CAGE)
3. CAGE outperforms best baseline by 10% and produces explanations to justify its predictions
4. Explanation transfer on two out-of-domain datasets

Related Work: Commonsense reasoning

- Commonsense reasoning datasets:
 - Story Cloze: predicting story ending from a set of plausible endings
 - Situations with Adversarial Generations (SWAG): predicting next scene based on initial event
- Models achieve human-level performance on some datasets
- Models struggle with understanding how pronouns resolve between sentences and world knowledge
- CQA addresses this by requiring models to infer from the question
- Language models perform poorly compared to human participants on CQA

Unclear: Do models actually do common-sense reasoning?

Related Work: Natural language explanations

- Rationale generation by highlighting complete phrases in input text that are sufficient to predict desired output (Lei et al., 2016)
- Human-generated natural language explanations to train a semantic parser to generate noisy labeled data and train a classifier for generating explanations (Hancock et al., 2018)
- Interpretability comes at the cost of loss in performance on Stanford Natural Language Inference dataset (Camburu et al., 2018)
- Multi-modal: Ensemble explanations and visual explanations improve performance (Rajani and Mooney, 2018 and 2017)

Do explanations for CQA lead to improved performance?

Related Work: Knowledge transfer in NLP

- Reliance on transfer of knowledge through pre-trained word vectors (e.g. Word2vec, GloVe) and contextualized word vectors (more refined with general encoding)
- Language models trained from scratch on large amounts of data and fine-tuned on specific tasks perform well
 - Only a few parameters need to be learned from scratch
 - Perform well on small amounts of supervised data
- **Gap: Fine-tuned language models don't perform as well on CQA**

Can we leverage these models to generate explanations and show that these explanations capture common sense?

Common Sense Explanations (CoS-E)

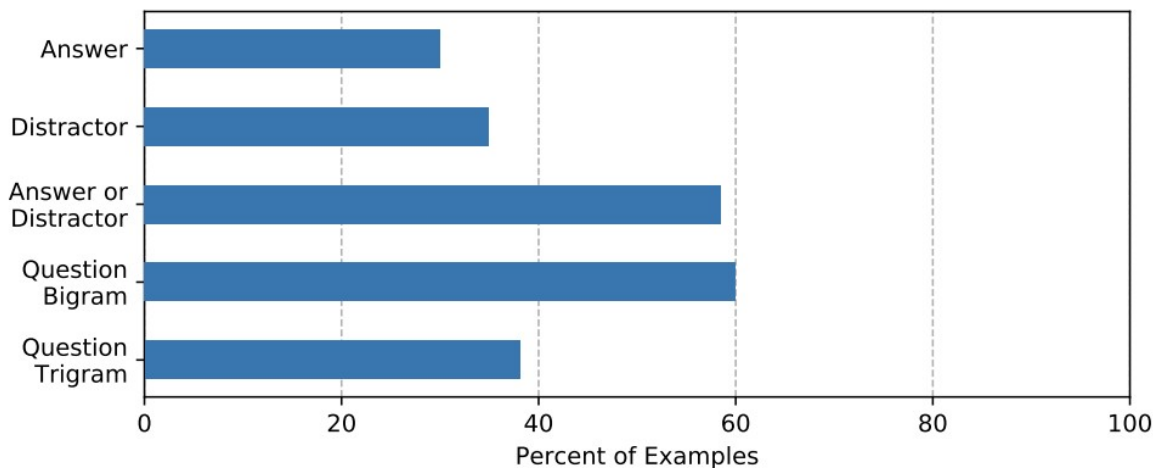
Dataset Structure/Creation

- Based on the CQA dataset
- CoS-E provides natural-language explanations for the correct answer choice and highlights important words in the question
- Explanations generated using MTurk
- Goal: To show whether models are performing reasoning correctly

Question:	While eating a hamburger with friends , what are people trying to do?	CQA
Choices:	have fun , tasty, or indigestion	
CoS-E:	Usually a hamburger with friends indicates a good time.	CoS-E

Dataset Considerations

- *CoS-E-selected* refers to the highlighted words, *CoS-E-open-ended* refers to the explanations
- Quality control was performed on the annotations and explanations
- Even explanations that don't discuss the ground truth answer are useful

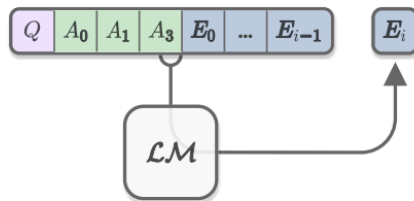


Commonsense Auto-Generated Explanations (CAGE)

CAGE Phase 1

Commonsense Auto-Generated Explanations (CAGE) Phase 1:

- Provide CQA example alongside corresponding CoS-E explanation to a language model
- Train model to generate the CoS-E explanation

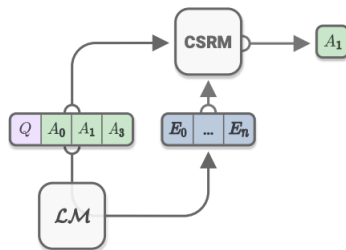


(a) One time-step of training a CAGE language model to generate explanations from CoS-E. It is conditioned on the question tokens Q concatenated with the answer choice tokens A_1, A_2, A_3 and previously generated tokens $E_1, \dots, E_{i-1}$. It is trained to generate token E_i .

CAGE Phase 2

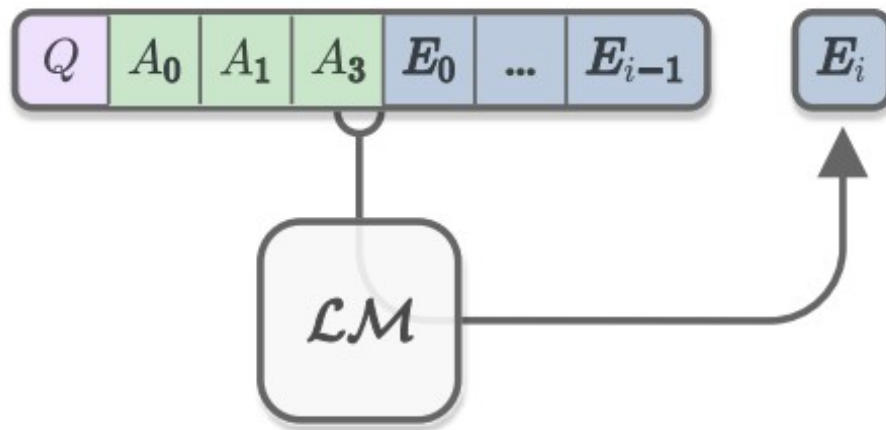
Commonsense Auto-Generated Explanations (CAGE) Phase 2:

- Use language models to generate explanations for each example in the training and validation sets of CQA
- Provide CAGE explanations to a second model by concatenating it to the original input (question, answer choices, and language model output)



(b) A trained CAGE language model is used to generate explanations for a downstream commonsense reasoning model (CSRM), which itself predicts one of the answer choices.

CAGE Phase 1: Intuition



One training step for CAGE

How is CAGE trained?

- Language Model trained to generate explanations from question-answer choice pairs
- Use pretrained OpenAI GPT
- Fine-tuned on the CQA and CoS-E dataset combination
- Two possible settings: explain-then-predict (**reasoning**) and predict-then-explain (**rationalization**)

CAGE Notation

- Question q
- Answer choices c_0, c_1, c_2
- Correct answer $a \in c_0, c_1, c_2$
- CoS-E explanation e_h
- CAGE predict explanation e

Reasoning

- Model is fine-tuned on the question, answer choices, and explanation tokens, but not the actual label.

$C_{RE} = “q, c0, c1, \text{ or } c2? \text{ commonsense says}”$

- Objective Function (canonical conditional language modeling objective):

$$\sum_i \log P(e_i | e_{i-k}, \dots, e_{i-1}, C_{RE}; \Theta)$$

Rationalization

- Model is now also given the ground truth label a :

$$C_{RA} = \text{“ } q, c_0, c_1, \text{ or } c_2? \text{ } a \text{ because ”}$$

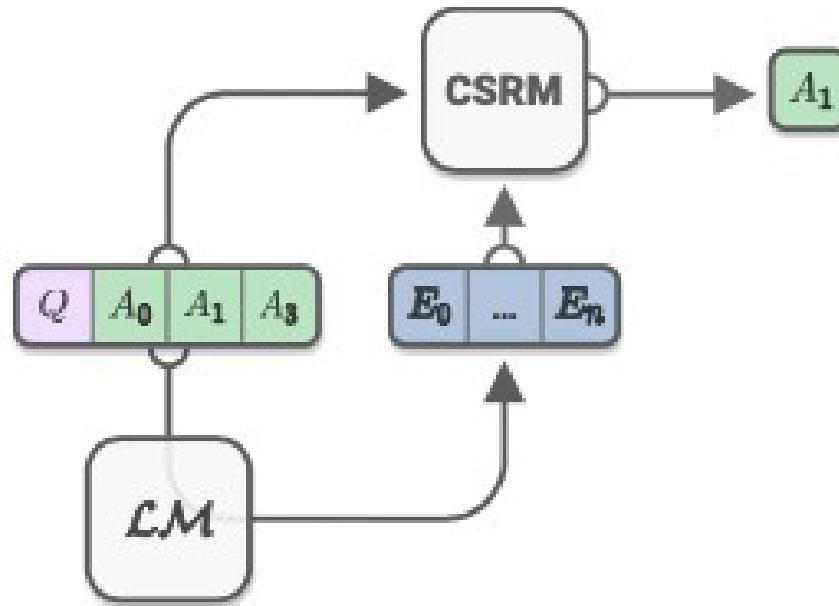
- Objective Function is the same as before but is also conditioned on the *label* a
- Thus, the explanations create rationalization that makes the model more interpretable

Training Parameters

- Generate sequences of maximum length 20
- Batch Size: 36, Epochs: 10
- Selected the best model using BLEU and perplexity scores

Commonsense Predictions with Explanations

CAGE Phase 2 Intuition



CAGE Inference

- Given human explanation from CoS-E or LM reasoning, can then perform predictions on CQA
- Simply concatenate Question, [Sep], Explanation, [Sep], Answer Choice as input to downstream CSRM (classifier)
- Use binary classification head on top of BERT backbone
 - 3 answer choices → 3 input sequences
 - Take sequence yielding highest confidence as output

CSRM (BERT) Training HP

- Train batch size: 24
- Test batch size: 12
- 10 training epochs
- Max sequence length of 50 for labels-only; 175 including explanations

Experimental Results

Experimental Results

Method	Accuracy (%)
BERT (baseline)	63.8
CoS-E-open-ended	65.5
CAGE-reasoning	72.6

Table 2: Results on CQA dev-random-split with CoS-E used during training.

Experimental Results

- Google search “question + answer choice” for each example and collected 100 top snippets per answer as context for **Reading Comprehension model**
- Extra data did not improve accuracy
- CAGE-reasoning resulted in **10% accuracy gain** over previous SOTA

Method	Accuracy (%)
RC (Talmor et al., 2019)	47.7
GPT (Talmor et al., 2019)	54.8
CoS-E-open-ended	60.2
CAGE-reasoning	64.7

Experimental Results

Method	Accuracy (%)
CoS-E-selected w/o ques	53.0
CoS-E-limited-open-ended	67.6
CoS-E-selected	70.0
CoS-E-open-ended w/o ques	84.5
CoS-E-open-ended*	89.8

Table 4: Oracle results on CQA dev-random-split using

- Oracle upper-bound: **human-generated explanations** from CoS-E provided during training and validation
- Unfair setting bc human that provided explanation had **ground truth answer**
- “**CoS-E selected**”: explanation consists of words humans selected as justification for model

Transferring Explanations Across Domains

- How well do natural language explanations transfer from CQA to SWAG and Story Cloze Test?
- Use GPT CAGE model fine-tuned on CQA train/dev to generate explanations on SWAG and Story Cloze Spring 2016 train/val
- Rinse and repeat using BERT with classifier head

Experimental Results

Method	SWAG	Story Cloze
BERT	84.2	89.8
+ expl transfer	83.6	89.5

- Camburu et al (2018) show that transferring explanations from SNLI to MultiNLI performs very poorly
- Transfer of explanations on commonsense reasoning tasks
- NLI problem has small fixed set of pre-defined labels unlike commonsense reasoning tasks like CQA, SWAG, Story Cloze
- **Adding explanations led to very small decrease in performance**

Qualitative Analysis

Analysis of CAGE

- CAGE-reasoning at train + validation → 72% accuracy
- CoS-E-open-ended performance at 90%—why the gap?
- Measure quality of CAGE
 - Human evaluation (42% CAGE vs 52% CoS-E-open-ended)
 - BLEU score measures syntactical precision by n-gram overlap
 - Perplexity: token-level measure of how well language models predict next word
- Result: beneficial to fine-tune the LM, but humans and LMs have widely varying ways of providing useful explanations

Analysis of baseline BERT model

- Error analysis on baseline BERT w/o explanations: performs poorly on longer / more compositional questions → explanations help w this
- CAGE reasoning typically simpler construction than CoS-E-open-ended, but adds meaningful context
- However, CAGE still provides “incorrect” answers often

Question: A child wants to play, what would they likely want?
Choices: **play tag**, breathe, fall down
CoS-E: A child to play tag
Reason Children want to play tag, and they want to play tag with their friends.
Rationale: Children want to play tag, what would they want to do?

Question:	What is the main purpose of having a bath?
Choices:	cleanness , use water, exfoliation, <u>hygiene</u> , wetness
Explanation:	the only purpose of having a bath is to clean yourself.

Question:	Where can you store you spare linens near your socks?
Choices:	cabinet, <u>chest</u> , hospital, dresser drawers , home
Explanation:	dresser drawer is the only place that you can store linens.

Question:	Where do you find the most amount of leafs?,
Choices:	forrest , floral arrangement, <u>compost pile</u> , field, ground
Explanation:	the most likely place to find leafs is in a garden.

Domain transfer: SWAG + Story Cloze

SWAG

Question:	Men are standing on motorbikes getting ready for a motocross competition.
Choices:	man places the ladders onto a fence and winds up a marching wall, high with hammer and a stone., man is talking to the camera and standing on a podium. , man stands outside in the field going at arms of people and leading a long jumping calf in front., man drops the javelin to the ground and jumps it very high.
Explanation:	man is talking to the camera and not the crowd.
Question:	The man examines the instrument in his hand.
Choices:	The person studies a picture of the man playing the violin., The person holds up the violin to his chin and gets ready. , The person stops to speak to the camera again., The person puts his arm around the man and backs away.
Explanation:	the person is holding the instrument in his hand.
Question:	The woman is seated facing the camera while another woman styles her hair.
Choices:	The woman in purple is wearing a blue dress and blue headband, using the pits to style her hair., The woman begins to cut the hair with her hair then serves it and begins brushing her hair and styling it., The woman puts some right braids on his., The woman continues to have her hair styled while turned away from the camera.
Explanation:	the woman is using the braids to trim her hair.

Story Cloze (ROCStories)

Question:	My friends all love to go to the club to dance. They think it's a lot of fun and always invite. I finally decided to tag along last Saturday. I danced terribly and broke a friend's toe.
Choices:	My friends decided to keep inviting me out as I am so much fun., The next weekend, I was asked to please stay home.
Explanation:	the next weekend, i would be asked to stay home
Question:	Ari spends \$20 a day on pickles. He decides to make his own to save money. He puts the pickles in brine. Ari waits 2 weeks for his pickles to get sour.
Choices:	Ari opens the jar to find perfect pickles. , Ari's pickles are sweet.
Explanation:	pickles are the only thing that can be found in a jar.
Question:	Gina sat on her grandpa's bed staring outside. It was winter and his garden was dead until spring. Her grandpa had passed away so there would be no one to tend it. The weeds would take over and strangle the flowers.
Choices:	Gina asked her grandpa what kind of flowers he liked best., Gina decided to go outside and pick some of the weeds.
Explanation:	the weeds would take over and strangle the flowers.

Conclusion & Group Discussion

Conclusion

- CoS-E on top of CommonsenseQA
- CAGE framework → LM leverages explanations
- Classifier on top of explanations
- SOTA performance on a difficult commonsense reasoning task
- Opens further avenues for studying explanation as it relates to interpretable commonsense reasoning

Discussion questions

- BLEU and perplexity as measures of goodness?
 - Has been shown multiple times to correlate poorly with human judgement
- Joint training of explanation and label? Not one prior to other
- Can commonsense reasoning help general reasoning (e.g. mathematics, Fermi, counterfactual) in other domains?
- How to align human/machine explanations?
- Research merit of this work?
- Recent work (RLPrompt, AutoPrompt) has shown that optimal prompts for LMs (with respect to downstream task performance) are often times gibberish
 - What does this say about the validity of using SOTA LMs for explanations?
 - Can we regularize LM training to better align with human reasoning?

Interpreting Language Models with Contrastive Explanations

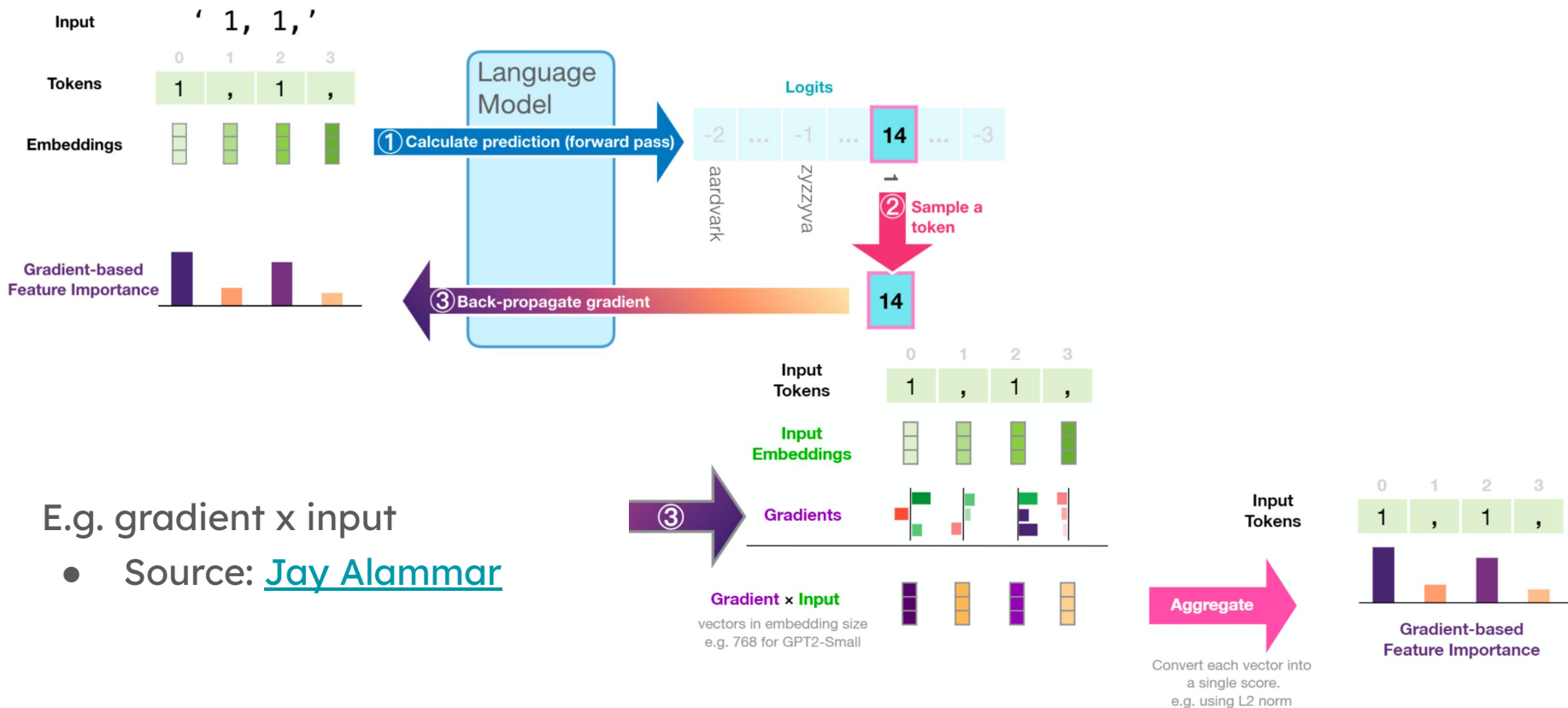
Kayo Yin & Graham Neubig

Presented by Charumathi Badrinath, Eric Shen, Leonard Tang, and Skyler Wu

Motivation + Example

- We've seen many interpretability and explanation strategies being applied to LMs, including transformer-based autoregressive LMs...
 - **Gradient-based/erasure-based feature attribution** methods provide a straightforward way to do this
 - Interpret each token of the input text as a feature
 - At each step, use gradients to calculate a **saliency score** to quantify the importance of each previous token to the model output (i.e. logit prediction for the next token)
- E.g. gradient x input

Motivation + Example



Motivation + Example

- **Problem:** For many LMs, using typical gradient- or erasure-based methods doesn't provide informative explanations
- Most of the time, the token with the highest saliency is the token immediately before the prediction
- How can we use these explanation ideas to create more meaningful attributions? By **contrasting** them with other token predictions.

Input: Can you stop the dog from
Output: barking
1. Why did the model predict "barking"?
Can you stop the dog from

Knowing the previous word is certainly very important for figuring out the next word, but that's not very helpful!

Main Contributions and Key Ideas

1. **Contrastive explanations:** why did the model predict one token *instead* of another? Which tokens were most influential? Extended previous methods.
2. **Grammatical consistency:** contrastive explanations > non-contrastive explanations w.r.t. verifying linguistic / grammatical phenomena.
3. **Human simulatability:** contrastive explanations help users better predict LLM behavior, also found to be more useful explanations by humans.

Non-contrastive gradient x input	{	Input: Can you stop the dog from
		Output: barking
Contrastive explanations	{	1. Why did the model predict “barking”?
		Can you stop the dog from
		2. Why did the model predict “barking” instead of “crying”?
	{	Can you stop the dog from
		3. Why did the model predict “barking” instead of “walking”?
	{	Can you stop the dog from

Red = raise probability of “barking,” Blue = decrease probability of “barking,” White = little influence.

GPT-2 ([Radford et al. 2019](#))

- Authors focus on GPT-2 (1.5B) and GPT-Neo (2.7B) → very similar to each other.
- **Training:** WebText dataset of 8 million web-pages.
 - No task-specific supervised training = “Multi-task training”
- **Objective:** predict the next word, given all previous words in input.
- **Behavior:** “chameleon-like,” adapts to style + content of the input text.
- **Architecture:** Transformer-based architecture.
 - “Autoregressive”: outputs tokens one at a time, *but* each token generated is appended to the input sequence → feed back to the model for next step.

Sources:

1. [OpenAI](#)
2. [Jay Alammar](#)
3. [Radford et al. 2018](#)

Gradient Norm Saliency Scores ([Simonyan et al. 2013](#))

- **Originally for image classification:** compute gradient of class score w.r.t input image, take the norm. Main idea = big gradient, big influence.
- **For LLMs:** compute gradient of next token in input sequence w.r.t. current input.

Specific token in input
(as embedding vector)

Entire input sequence (as
embedding)

$g(x_i) = \nabla_{x_i} q(y_t | \mathbf{x})$

g outputs a
vector!

Model output for
the predicted
token, given input
(logit)

→

$S_{GN}(x_i) = ||g(x_i)||_{L1}$

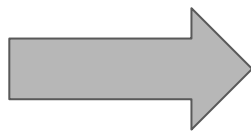
Saliency score

Gradient x Input Saliency Scores ([Shrikumar et al. 2016](#))

- **Method:** Similar gradient computation as Gradient Norm, simply replacing L1 norm with dot product with input itself.

Exact same gradient computation

$$g(x_i) = \nabla_{x_i} q(y_t | \mathbf{x})$$



Dot product, instead of L1 norm

$$S_{GI}(x_i) = g(x_i) \cdot x_i$$

Input Erasure Saliency Scores ([Li et al. 2016](#))

- **Intuition:** how does erasing different parts of the input affect the output?
- **Procedure:** compute difference in model outputs using full input vs. input with a specific token zeroed out. NOT gradient-based!

$$S_E(x_i) = \underbrace{q(y_t|\mathbf{x})}_{\text{Probability of output given input *without* token of interest}} - \underbrace{q(y_t|\mathbf{x}_{\neg i})}_{\text{Probability of output given entire input}}$$

Saliency score of token of interest

Related Work + Relevant Limitations (Pt. 1)

- Non-contrastive saliency score methods:
 - Simonyan et al. 2013, Shrikumar et al. 2016, Li et al. 2016 (see previous slides). Not NLP specific though!
 - Not very developed in NLP use cases.
 - When applied to NLP (e.g., Wallace et al. 2019), authors found that methods often **returns last token before output as most influential.**
- Adversarial Methods on NLP:
 - Wallace et al. 2019: HotFlip on NLPs, replace words to change model's prediction. AllenNLP suite.

Related Work + Relevant Limitations (Pt. 2)

- Counterfactual explanations in *text classification*:
 - Jacovi et al. 2021: erase features from input, project input representation into “contrastive space” → measure importance of erased feature by comparing class probabilities before / after erasure.
 - But, unsure how to extend into *language modeling* space, with much bigger input + output spaces.
- Contrastive methods are not new, just not used for NLP very much (Stepin et al. 2021, survey).

Method

- Simple modification to formulation of existing gradient-based explanations.
- **Contrastive** setup:

“Given input token (embedding) sequence $\mathbf{x}$, why did the model predict this next token y ?”

Given $\mathbf{x}$, why did the model predict this next **target token** y_t **instead of a foil token** y_f ?

Calculate gradients of the logit of y with respect to input tokens

Calculate gradients of the **difference in logits** between y_t and y_f with respect to input tokens

Method

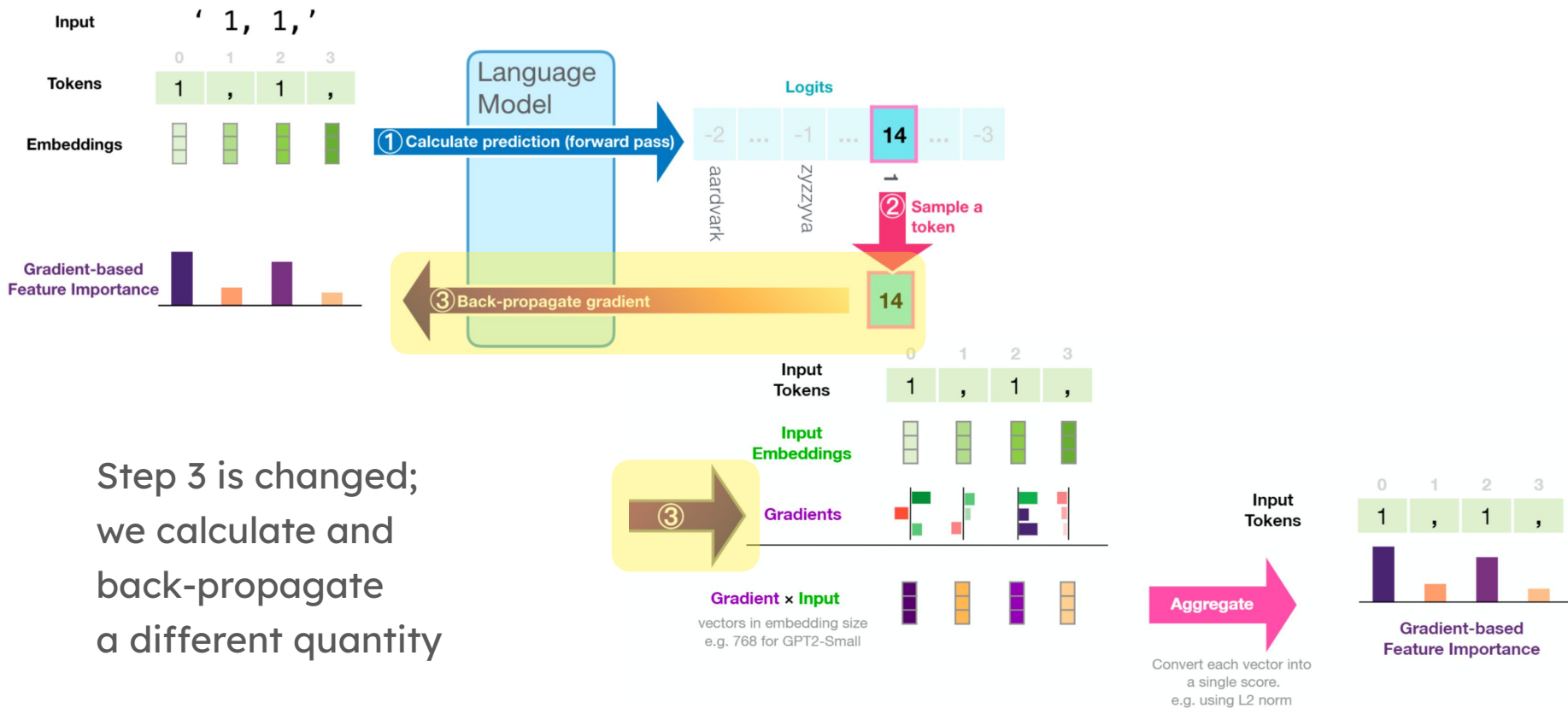
Let $q(y_t | \mathbf{x})$ be the model output for token y_t given the input $\mathbf{x}$.

Let $S(x_i)$ be the saliency score for token x_i in input $\mathbf{x}$.

Let $\mathbf{x}_{-i}$ be the input $\mathbf{x}$ where x_i is zeroed out.

Saliency Score Explanation Method	Non-Contrastive (Standard) Saliency Calculation	Contrastive Saliency Calculation
Gradient Norm	$S_{GN}(x_i) = \ \nabla_{x_i} q(y_t \mathbf{x})\ _{L_1}$	$S_{GN}^*(x_i) = \ \nabla_{x_i} (q(y_t \mathbf{x}) - q(y_f \mathbf{x}))\ _{L_1}$
Gradient x Input	$S_{GI}(x_i) = \nabla_{x_i} q(y_t \mathbf{x}) \cdot x_i$	$S_{GI}(x_i) = \nabla_{x_i} (q(y_t \mathbf{x}) - q(y_f \mathbf{x})) \cdot x_i$
Input Erasure	$S_E(x_i) = q(y_t \mathbf{x}) - q(y_t \mathbf{x}_{-i})$	$S_E^*(x_i) = (q(y_t \mathbf{x}) - q(y_t \mathbf{x}_{-i})) - (q(y_t \mathbf{x}_{-i}) - q(y_f \mathbf{x}_{-i}))$

Method



Do Contrastive Explanations Identify Linguistically Appropriate Evidence?

Q: Are contrastive explanations >> non-contrastive explanations in identifying words that we think should influence the output token?

Experimental Setup:

- **BLiMP dataset:** pairs of minimally different English sentences that contrast in grammatical acceptability under some linguistic paradigm
- 5 linguistic phenomena with 12 paradigms + used spaCy NLP library to extract grammatically relevant parts of each sentence

Do Contrastive Explanations Identify Linguistically Appropriate Evidence?

gender + number of pronoun must agree with antecedent

action verbs used with animate objects

some negative polarity words only appear in some contexts

number of subject and verb in present tense must agree

Phenomenon	UID ³	Acceptable Example	Unacceptable Example
Anaphor Agreement	aga ana	<u>Katherine</u> can't help herself . Many <u>teenagers</u> were helping themselves .	<u>Katherine</u> can't help himself . Many <u>teenagers</u> were helping herself .
Argument Structure	asp	Amanda was <u>respected</u> by some waitresses .	Amanda was <u>respected</u> by some picture .
Determiner-Noun Agreement	dna dnai dnaa dnaai	Craig explored <u>that</u> grocery store . Phillip was lifting <u>this</u> mouse . Tracy praises those lucky guys . This person shouldn't criticize <u>this</u> upset child .	Craig explored <u>that</u> grocery stores . Phillip was lifting <u>this</u> mice . Tracy praises <u>those</u> lucky guy . This person shouldn't criticize <u>this</u> upset children .
NPI Licensing	npi	<u>Even</u> these trucks have often slowed.	<u>Even</u> these trucks have ever slowed.
Subject-Verb Agreement	darn ipsv rpsv	A sketch of lights doesn't appear. This goose isn't bothering Edward. Jeffrey hasn't criticized Donald.	A sketch of lights don't appear. This goose weren't bothering Edward. Jeffrey haven't criticized Donald.

Do Contrastive Explanations Identify Linguistically Appropriate Evidence?

Evaluation Metrics:

S = explanation vector; S_i = saliency of x_i

E = known evidence; $E_i = 1$ (x_i grammatically influences output token)

- $S \cdot E$ → sum of saliency scores of all input tokens that are part of evidence
- Probes needed → ranking of first token x_i where $E_i = 1$ when sorted by decreasing saliency
- MRR (mean reciprocal rank) → average (over all sentences) of inverse rank of first x_i where $E_i = 1$ when sorted in descending saliency

Findings: Linguistic Agreement

- Contrastive (cf. non-contrastive) explanations are more aligned with linguistic paradigms
- Contrastive explanations have a better alignment with BLiMP than random vectors baseline
 - Note high baseline
- Non-contrastive explanations do not outperform random baseline

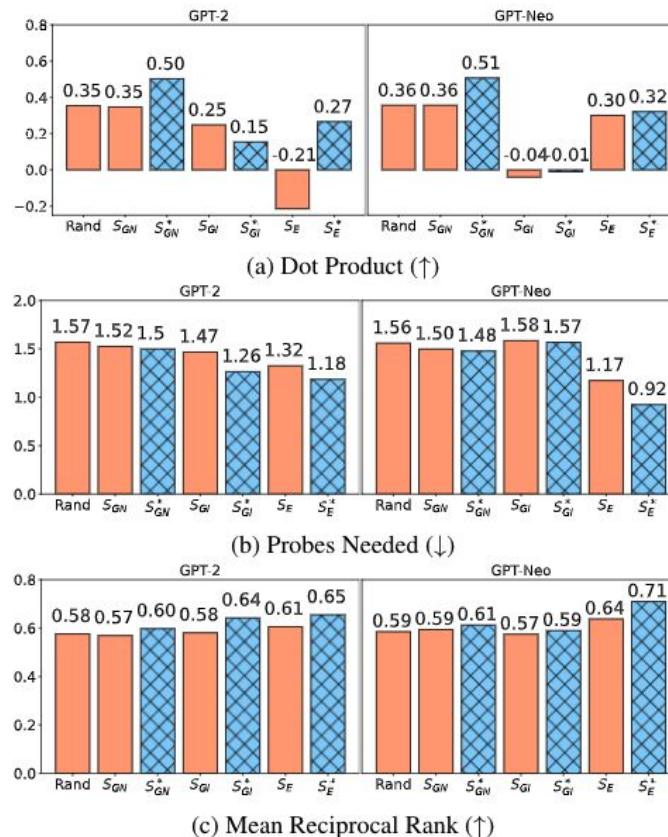


Figure 1: Alignment of different GPT-2 (left) and GPT-Neo (right) explanations with known evidence in BLiMP according to dot product (top), probes needed (middle), mean reciprocal rank (bottom) averaged over linguistic paradigms.

Findings: Linguistic Agreement

- Further apart *known evidence* token is from *target token* $\rightarrow$ larger increase in alignment (MRR) of contrastive cf. non-contrastive
- Contrastive explanations can particularly capture model decisions requiring *longer-range context*

Katherine can't help **herself**.
Phillip was lifting this **mouse**.

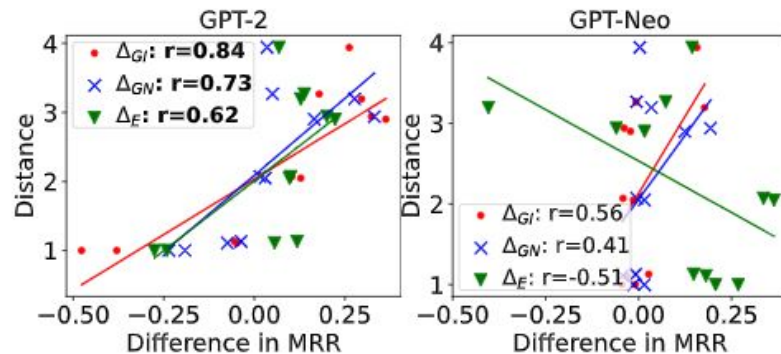


Figure 2: Scatter plot of the average distance of the known evidence to the target token across each linguistic paradigm against the difference in MRR scores between the contrastive and non-contrastive versions of each explanation method, with the Pearson correlation for each explanation method. Statistically significant Pearson's r values ($p < 0.05$) are in **bold**. In most cases, there is a positive correlation between the increase in MRR and the distance of the evidence.

Do Contrastive Explanations Help Users Predict LM Behavior?

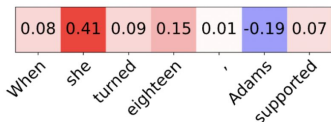
Q: Do contrastive explanations increase users' ability to predict a model's output token (i.e. "simulate" model behavior)?

** normalized for model accuracy, # of each condition

Experimental Setup:

model = GPT-2

explanation = {no explanation, gradient x input, contrastive gradient x input, erasure, contrastive erasure}



Which token did the model more likely predict?

☐ herself

☐ himself

Was the explanation useful in making your decision?

☒ Yes

☐ No

Correct!

10 word pairs from BLiMP, 10 word pairs selected to maximize confusion score on WikiText-103 test split

$$P(x_{true} = a, x_{model} = b) =$$

$$\frac{1}{N} \sum_{x \in X} \sum_{t \in \text{pos}(x) | x_t = a} P_{model}(\hat{x}_t = b | x_{<t})$$

$$\mathcal{C}(a, b) = \min(P(x_{true} = a, x_{model} = b), P(x_{true} = b, x_{model} = a))$$

Figure 3: Example of a prompt in our human study.

Findings: User Alignment

- All four types of explanations help users simulate model behavior
- *Contrastive explanations* lead to *more accurate simulations*
- *Contrastive explanations* are considered *more useful*
- Takeaway: contrastive explanations help human observers accurately simulate model predictions the most

	Acc.	Acc. Correct	Acc. Incorrect	Useful	Acc. Useful	Acc. Not Useful
None	61.38	74.50	48.25	–	–	–
S_{GI}	64.00	78.25	49.75	62.12	67.20	58.75
S_{GI}^*	65.62	79.00	52.25	63.88	69.67	58.48
S_E	63.12	79.00	47.25	46.50	65.86	60.75
S_E^*	64.62	77.00	52.25	64.88	70.52	53.74

Table 3: Simulation accuracy (%) in predicting GPT-2 outputs and subjective usefulness of explanations for various explanation methods. For each explanation method, scores that are statistically significantly higher ($p \leq 0.05$) than the analogous method with a different contrastive setting are bolded. Overall, users achieve higher simulation accuracy with contrastive explanations.

What Context Do Models Use for Certain Decisions?

Q: How do language models achieve various linguistic distinctions? Is similar evidence necessary to disambiguate foils that are similar linguistically?

Experimental Setup:

Targets = 10 most frequent words in each major part of speech

Foils = 10000 most frequent vocab items

- For each target y_t select 500 sentences from WikiText-103 $\rightarrow$ “sentence set” X
- For each foil y_f and each sentence x_i in X generate contrastive explanation $e(x_i, y_t, y_f)$ + concatenate them
- For each target y_t apply k-means on $e(x_i, y_t, \text{all foils})$

Idea: Explanation vectors represent *type of context* needed to disambiguate foil from target; want to see if clusters associated with salient linguistic distinctions

Findings: What Context Do Models Use For Decisions?

- (Paradigmatically) linguistically similar foils cluster together
 - Foil clusters reflect linguistic distinctions unique from word embeddings
- Model use similar types of input features to make certain decisions
 - Animacy Ex) Target is animate, foils in cluster are all inanimate
- Examining cluster explanations (maps) yields insights into GPT-2 (“BERTology”)
 - Pronoun Ex) GPT-2 influenced by unrelated pronouns → produces incorrect gender

Phenomenon / POS	Target	Foil Cluster	Embd Nearest Neighbors	Example
Anaphor Agreement	he	<u>she</u> , her, She, Her, herself, hers	she , She, her , She , he, they, Her , we, it, she, I, that, Her, you, was, there, He, is, as, in'	That night , Ilisa confronts Rick in the deserted café . When he refuses to give her the letters , _____
Animate Subject	man	<u>fruit</u> , mouse, ship, acid, glass, water, tree, honey, sea, ice, smoke, wood, rock, sugar, sand, cherry, dirt, fish, wind, snow	fruit , fruits, Fruit, meat, flower, fruit, tomato, vegetables, fish , apple, berries, food, citrus, banana, vegetable, strawberry, fru, delicious, juice, foods	You may not be surprised to learn that Kelly Pool was neither invented by a _____
Determiner-Noun Agreement	page	<u>tabs</u> , <u>pages</u> , icons, stops, boxes, doors, shortcuts, bags, flavours, locks, teeth, ears, tastes, permissions, stairs, tickets, touches, cages, saves, suburbs	tabs , tab, Tab, apps, files, bags, tags, websites, sections, browsers, browser, icons, buttons, pages , keeps, clips, updates, 28, insists, 14	Immediately after "Heavy Competition" first aired, NBC created a sub- _____

Strengths/Weaknesses

A cursory list:

- Simple yet effective modification applicable to a variety of feature attribution methods using saliency scores
- Easy to compute, extensible to general LMs (including for NMT)
- Evaluated with interesting foil clustering analysis
- Empirically shown to help with human observers (empirically good for interpretability)
- Only applied to three feature attribution methods in the paper
- Only GPT-2 and GPT-Neo used as LM examples in main papers
- Human study very limited in scope
- Does not attempt to look at model internals; saliency scores are arguably a crude approximation of interpretation

Questions for the Audience

- Given that LLMs (and other ML models) often exhibit “phase shifts” at different sizes to what extent do you expect the results from this paper to generalize to cutting-edge models like GPT-4?
- In practice, how would one create foils for free-response questions and/or general conversational use? How generalizable are these contrastive tools?
- How effective do you think saliency scores (through gradient/erasure-base explanation methods, etc.) are for achieving interpretability?
- How much do we trust the GPT-2 embeddings (this is the primary workhorse for most of their methods) and the generalizability of the authors’ results?